

Study design: a multi-tissue MRI aging clock and age acceleration

Cohort & segmentation

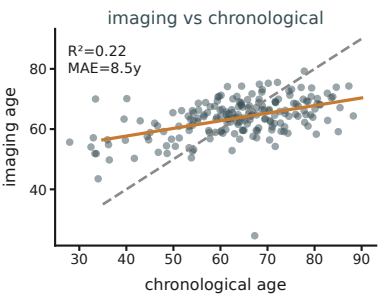
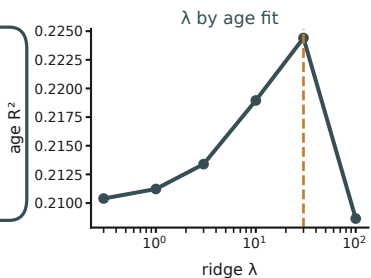
204 patients · preoperative lumbar MRI
Automated multi-tissue segmentation
(3D Slicer / TotalSegmentator)

Scanner-robust features:
size-normalized volumes + vertebra-referenced
T2 intensity ratios (iliopsoas, deep paraspinal,
vertebra, disc, cord)

[segmented MRI panel]

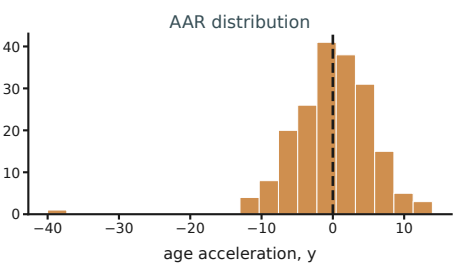
MRI aging clock

Ridge regression predicts
chronological age from the
multi-tissue signature
(10-fold CV; λ by age R^2)



Age acceleration

Age acceleration (AAR) = residual of
imaging age $\sim$ chronological age
Orthogonal to chronological age (no suppression)
positive = tissues look older than the patient's age



Association analysis

Firth penalized logistic regression
Non-home discharge $\sim$ age acceleration
+ chronological age + sex + ASA class
Sensitivity: clock specification $\cdot \lambda \cdot$ seed

AAR (per 1-SD)	No.	OR (95% CI)	P Value
Crude	192	1.85 (1.25-2.83)	.002
Adjusted for age	192	2.00 (1.32-3.14)	<.001
Adjusted for age, sex, ASA	192	1.84 (1.18-2.95)	.007

